Type Theory & Categorical Models: A Unified Approach

From Typelis to Logic to Closed Cartesian Categories

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Assumptions

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Syntax of -terms

- Variables, abstraction $\lambda x.M$, application MN

Subterms & Free variables

- Subterm relation: $N \preceq M$ if N occurs in M
- Free variables $\mathsf{FV}(M)$
- Combinators, when $\mathsf{FV}(M) = \varnothing$

Substitution

- Capture-avoiding substitution $M[x := N]$

α-equivalence

- Renaming of bound variables: $\lambda x.M \equiv_\alpha \lambda y.M[x := y]$ if $y \notin FV(M)$
- Formal inference rules: reflexivity, congruence, renaming under binder

β-reduction

- Single-step: $(\lambda x.M)N \rightsquigarrow_\beta M[x := N]$
- Multi-step: $\rightsquigarrow_\beta^*$ (reflexive, transitive closure)
- Redex = reducible expression; normal form = no redex

η-equivalence

- Extensionality: $\lambda x.M\,x =_\eta M$ if $x \notin FV(M)$

Church–Rosser

- Confluence of β-reduction:

$$\mathrm{M} \rightsquigarrow_\beta^* \mathrm{N}_1 \text{ and } \mathrm{M} \rightsquigarrow_\beta^* \mathrm{N}_2 \text{ then } \exists \mathrm{P}.\ \mathrm{N}_1 \rightsquigarrow_\beta^* \mathrm{P} \wedge \mathrm{N}_2 \rightsquigarrow_\beta^* \mathrm{P}$$

- Proof idea: parallel reduction / diamond property

Normalization & Examples

- Strongly normalizing, weakly normalizing, non-normalizing
- λx.x (normal), Ω (non-normalizing), $(\lambda \mathrm{x.y})\Omega$ (reduces to y)

Encodings & Fixed-points

- Church numerals: $\mathsf{ZERO}, \mathsf{SUCC}, \mathsf{ADD}, \mathsf{MUL}, \mathsf{PRED}$
- Booleans: TRUE, FALSE, IF
- Pairs: PAIR, FST, SND
- Lists: NIL, CONS, HEAD, TAIL
- Fixed-point combinators: $\mathrm{Y}, \mathrm{Z}, \Theta$

Typing

Minimal Typing

Types

$$\mathscr{T} ::= \mathscr{B} \mid \mathscr{T} \to \mathscr{T}$$

$$\lambda \mathrm{x} : \sigma . \mathrm{x} \vdash \lambda \mathrm{x} : \sigma . \mathrm{x} : \sigma \to \sigma$$

$$\lambda \mathrm{x} . \mathrm{x} \vdash \lambda \mathrm{x} . \mathrm{x} : \sigma \to \sigma$$

Church–Rosser ✓ Strongly Normalizing ✓ Weakly Normalizing ✓

Type Extensions

Product Types:	$A \times B \to A$	$A \times B \to B$
Sum Types:	$A \to A + B$	$B \to A + B$

In terms of β-reduction they are all confluent.

```
-- Product Type: A x B
data Pair a b = Pair a b
-- Projections
pi1 :: Pair a b -> a
pi1 (Pair x _) = x
pi2 :: Pair a b -> b
pi2 (Pair _ y) = y
```

```
-- Sum type: A + B
data Sum a b = InL a | InR b
-- Injections
inl :: a -> Sum a b
inl = InL
inr :: b -> Sum a b
inr = InR
```

Extended Typing – Products

Product Types

$$\mathscr{T}^{\times} ::= \mathscr{B} \mid \mathscr{T}^{\times} \to \mathscr{T}^{\times} \mid \mathscr{T}^{\times} \times \mathscr{T}^{\times} \mid 1$$

Product Terms

$$\Lambda^{\times} ::= x \mid \lambda x : \mathscr{T}.\ \Lambda^{\times} \mid \Lambda^{\times}\ \Lambda^{\times} \mid \langle \Lambda^{\times}, \Lambda^{\times} \rangle \mid \pi_1 \Lambda^{\times} \mid \pi_2 \Lambda^{\times} \mid *$$

Extended Typing – Sums

Sum Types

$$\mathscr{T}^+ ::= \mathscr{B} \mid \mathscr{T}^+ \to \mathscr{T}^+ \mid \mathscr{T}^+ + \mathscr{T}^+ \mid 0$$

Sum Terms

$$\Lambda^+ ::= \mathrm{x} \mid \lambda \mathrm{x} : \mathscr{T} .\ \Lambda^+ \mid \Lambda^+\,\Lambda^+ \mid \iota_1 \Lambda^+ \mid \iota_2 \Lambda^+ \mid \mathrm{case}\ \Lambda^+ \ \mathrm{of}\ \iota_1 \mathrm{x} \Rightarrow \Lambda^+ \mid \iota_2 \mathrm{y} \Rightarrow \Lambda^+ \mid \Box_{\mathrm{A}}\,\Lambda^+$$

Extended Typing – Products and Sums

Product Sum Types

$$\mathscr{T}^{\times+} ::= \mathscr{B} \mid \mathscr{T}^{\times+} \to \mathscr{T}^{\times+} \mid \mathscr{T}^{\times+} \times \mathscr{T}^{\times+} \mid \mathscr{T}^{\times+} + \mathscr{T}^{\times+} \mid 1 \mid 0$$

Product Sum Terms

$$\Lambda^{\times+} ::= \Lambda^{\times} \cup \Lambda^{+}$$

- Church-Rosser Property (Untyped λ-Calculus):
  - Confluence of reduction: if $M \rightsquigarrow_\beta^* N_1$ and $M \rightsquigarrow_\beta^* N_2$, there exists P with $N_1 \rightsquigarrow_\beta^* P$ and $N_2 \rightsquigarrow_\beta^* P$
  - Proven using parallel $\beta\eta$-reduction
- Weak Normalization (Typed λ-Calculus with Products):
  - Every typable term eventually reduces to a normal form
  - Degree is used as a measure of term complexity to structure the induction
  - Extends standard normalization proofs to include product types

Curry Howard

Intuitionistic Logic

- Proofs must be constructive:
  - To prove $A \to B$, one must give a method that transforms any proof of A into a proof of B.
  - To prove $\exists x. P(x)$, one must provide a concrete witness a together with a proof of $P(a)$.
- Excluded Middle does not hold in general: $\nvdash A \vee \neg A$ unless one can actually prove one of the disjuncts.
- Proofs by contradiction are not allowed.

The Isomorphism

Intuitionistic Logic	Typed λ-Calculus
Proposition A	Type A
Proof of A	Term of type A
Implication $A \to B$	Function type $A \to B$
Conjunction $A \wedge B$	Product type $A \times B$
Disjunction $A \vee B$	Sum type $A + B$
$\top$	1
$\bot$	0
Proof normalization	β-reduction

Example: Modus Ponens

Modus Ponens

$$\frac{P \to Q \quad P}{Q}\,\mathrm{MP}$$

$$\begin{aligned}
&(\lambda f : P \to Q.\lambda x : P.f\,x)\ f : P \to Q\ x : P && \vdash (P \to Q) \to P \to Q \\
&\rightsquigarrow_\beta\ (\lambda x : P.f\,x)\ x : P && \vdash P \to Q \\
&\rightsquigarrow_\beta\ f\,x && \vdash Q
\end{aligned}$$

Closed Cartesian Categories

What is Category Theory

- Category theory is a general theory of mathematical structures and their relations
- The focus of study are relations among objects, and not the objects themselves

Definition of Category

Category

A category $\mathscr{C}$ consists of:

- A collection of objects: $\mathsf{Ob}(\mathscr{C})$
- For each pair of objects $A, B \in \mathsf{Ob}(\mathscr{C})$, a set of arrows $\mathsf{Hom}_{\mathscr{C}}(A, B)$
- For each $A \in \mathsf{Ob}(\mathscr{C})$, an identity morphism $\mathsf{id}_A \in \mathsf{Hom}_{\mathscr{C}}(A, A)$
- For each $f \in \mathsf{Hom}_{\mathscr{C}}(A, B)$ and $g \in \mathsf{Hom}_{\mathscr{C}}(B, C)$, a composition $g \circ f \in \mathsf{Hom}_{\mathscr{C}}(A, C)$

Axioms

These must satisfy:

- Associativity: $(h \circ g) \circ f = h \circ (g \circ f)$.
- Identity: $\mathsf{id}_B \circ f = f = f \circ \mathsf{id}_A$.

Example: Preorder as a Category

$$
\begin{tikzcd}
a \arrow[r, "f"] \arrow[rr, bend left=50, "g \circ f"] \arrow[rrr, bend left=60, "(h \circ g) \circ f = h \circ (g \circ f) = h \circ g \circ f"] & b \arrow[r, "g"] \arrow[rr, bend right=50, "h \circ g"'] & c \arrow[r, "h"] & d
\end{tikzcd}
$$

Example: Preorder as a Category

$$
\begin{tikzcd}
a \arrow[r, "\leq"] \arrow[rr, bend left, "\leq"] \arrow[rrr, bend left=50, "\leq"] & b \arrow[r, "\leq"] \arrow[rr, bend right, "\leq"'] & c \arrow[r, "\leq"] & d
\end{tikzcd}
$$

Commutative Diagrams

Definition

A diagram in a category $\mathscr{C}$ is said to commute if, for any two objects connected by more than one path of morphisms, all paths compose to the same morphism.

$$\begin{tikzcd} A \arrow[r, "f"] \arrow[dr, "h"'] & B \arrow[d, "g"] \\ & C \end{tikzcd}$$

This triangle commutes if:

$$g \circ f = h$$

meaning the two ways of going from A to C are equal. Commutativity ensures consistency of compositions.

Injectivity via Arrows

Injective Function

A function $f : X \to Y$ is injective if $\forall x_1, x_2 \in X, f(x_1) = f(x_2) \implies x_1 = x_2$

Monomorphism

A morphism $f : X \to Y$ in a category $\mathscr{C}$ is a monomorphism if for all objects Z and morphisms $g_1, g_2 : Z \to X$,

$$f \circ g_1 = f \circ g_2 \implies g_1 = g_2.$$

- In the category Set, monomorphisms coincide with injective functions.
- This definition generalizes injectivity to arbitrary categories.

Universal Constructions

Definition

Universal Constructions are a way of defining properties up to Isomorphism using only arrows

Example of Universal Construction: Terminal Object

Definition

An object T in a category $\mathscr{C}$ is terminal if for every object $X \in \mathsf{Ob}(\mathscr{C})$ there exists a unique morphism

$$!_X : X \longrightarrow T$$

Take terminal objects $A, B \in \mathsf{Ob}(\mathscr{C})$. Then,

$$\begin{tikzcd}
b \arrow[r] \arrow[dr] & B \arrow[d, leftrightarrow] \\
c \arrow[r] \arrow[ur] & A \\
d \arrow[r] \arrow[ur] &
\end{tikzcd}$$

$$A \cong B$$

Products

$$A \xleftarrow[\pi_1]{} A \times B \xrightarrow[\pi_2]{} B$$

$$A = \{2,4,6\} \quad B = \{3,6,9\} \quad A \times B = \{(2,3),(2,6),(2,9),(4,3),\ldots\}$$

Products

$$\begin{tikzcd}
 & X \arrow[dl, "f"'] \arrow[dr, "g"] & \\
A & A \times B \arrow[l, "\pi_1"] \arrow[r, "\pi_2"'] & B
\end{tikzcd}$$

$$A = \{2,4,6\} \quad B = \{3,6,9\} \quad A \times B = \{(2,3),(2,6),(2,9),(4,3),\ldots\}$$

$$X = \{1,2,3\} \quad f(x) = 2x \quad g(x) = 3x$$

Products

$$\begin{tikzcd}
 & X \arrow[dl, "f"'] \arrow[d, dashed, "\langle f,g \rangle"] \arrow[dr, "g"] & \\
A & A \times B \arrow[l, "\pi_1"] \arrow[r, "\pi_2"'] & B
\end{tikzcd}$$

$$A = \{2,4,6\} \quad B = \{3,6,9\} \quad A \times B = \{(2,3),(2,6),(2,9),(4,3),\ldots\}$$

$$X = \{1,2,3\} \quad f(x) = 2x \quad g(x) = 3x$$

$$\langle f,g \rangle = \langle 2x, 3x \rangle$$

Products

Definition: Product

$$
\begin{tikzcd}
 & X \arrow[dl, "f"'] \arrow[d, dashed, "\langle f,g \rangle"] \arrow[dr, "g"] & \\
A & P \arrow[l, "\pi_1"] \arrow[r, "\pi_2"'] & B
\end{tikzcd}
$$

Duality in Category Theory

Definition: For any category $\mathscr{C}$, the dual category $\mathscr{C}^{\mathrm{op}}$ is defined by:

- $\mathrm{Ob}(\mathscr{C}^{\mathrm{op}}) = \mathrm{Ob}(\mathscr{C})$
- $\mathrm{Hom}_{\mathscr{C}^{\mathrm{op}}}(A, B) = \mathrm{Hom}_{\mathscr{C}}(B, A)$
- Composition in $\mathscr{C}^{\mathrm{op}}$ is reversed: if $f : B \to C$ and $g : A \to B$ in $\mathscr{C}$, then $g \circ f$ in $\mathscr{C}^{\mathrm{op}}$ corresponds to $f \circ g$

Intuition: Flip all arrows in the category.

Example: Duality of Products – Coproducts

$$
\begin{tikzcd}
 & X \arrow[dl, "f"'] \arrow[d, dashed, "\langle f,g \rangle"] \arrow[dr, "g"] & \\
A & A \times B \arrow[l, "\pi_1"] \arrow[r, "\pi_2"'] & B
\end{tikzcd}
\quad \Rightarrow \quad
\begin{tikzcd}
A \arrow[r, "\iota_1"] \arrow[dr, "f"'] & A + B \arrow[d, dashed, "{[f,g]}"] & B \arrow[l, "\iota_2"'] \arrow[dl, "g"] \\
 & X &
\end{tikzcd}
$$

Exponential Object

Definition: Exponential

$$\begin{tikzcd}
C \times A \arrow[rr, dashed, "\lambda f \times 1_A"] \arrow[dr, "f"'] & & B^A \times A \arrow[dl, "ev_{A,B}"] \\
& B &
\end{tikzcd}$$

- $f : C \times A \to B$ (uncurried) $\qquad f = \lambda(c,a).\, g(c,a)$
- $\lambda f : C \to B^A$ (curried) $\qquad \lambda f = \lambda c.\, (\lambda a.\, g(c,a))$
- $ev_{A,B} : B^A \times A \to B$ (evaluation) $\qquad f = ev_{A,B} \circ ((\lambda f) \times id_A)$

$$f = \lambda(a,b).\, g(a,b) \quad \rightsquigarrow \quad \lambda f = \lambda a.\, \lambda b.\, g(a,b)$$

Cartesian Closed Category

Definition: Closed Cartesian Category

A category $\mathscr{C}$ is cartesian closed if:

- $\mathscr{C}$ has a <u>terminal object 1</u>
- $\mathscr{C}$ has binary products <u>$A \times B$ for all objects A, B</u>
- $\mathscr{C}$ has exponentials: <u>for all objects A, B, there exists B^A</u> such that

$$\mathsf{Hom}_{\mathscr{C}}(X \times A, B) \cong \mathsf{Hom}_{\mathscr{C}}(X, B^A)$$

naturally in X

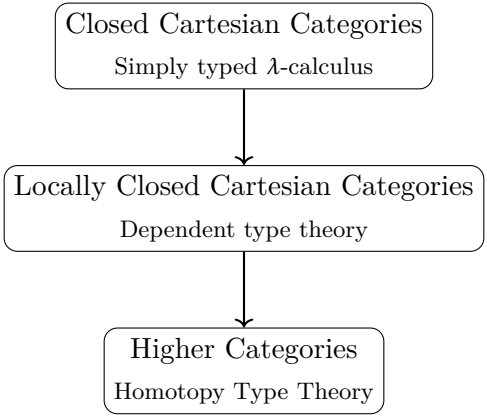
Intuition: Functions as objects product + exponent = function space.

The Lambek Extension

Intuitionistic Logic	Typed λ-Calculus	Category Theory / Lambek
Proposition A	Type A	Object A
Proof of A	Term of type A	Morphism $f : 1 \to A$
Implication $A \to B$	Function type $A \to B$	Exponential object B^A
Conjunction $A \wedge B$	Product type $A \times B$	Product $A \times B$
$\top$	1	Terminal object 1
Proof normalization	β-reduction	Commutativity of diagrams

Intuitionistic Logic	Typed λ-Calculus	Category Theory / Lambek
Disjunction $A \vee B$	Sum type $A + B$	Coproduct $A + B$
$\bot$	Empty type 0	Initial object 0

Perspectives



Errata

p.31 (Adgda -> Agda)

p.31 (Duplicated definition + Empty definition)

p.38 (Ax -> Assume-X)

p.48 (belements -> elements)

p. (Locally Cartesian Closed -> Locally Closed Cartesian)

Inconsistencies here and there + Formatting issues

Q&A.